# INFO 5368-030: Practical Applications in Machine Learning

**Due**: May 12th at 10:30AM
**Assignment Type**: Group (Up to 6 members)
**Submit Slides on**: Gradescope (Entry code: 7KB3DP); Submit PDF.
**Total Points**: 45 points
NO LATE SUBMISSIONS ALLOWED

CORNELL TECH

## Final Project Presentation

## Overview

In final project presentations, student teams will present projects for up to **8 minutes maximum** on May 12, 2026 from 10:30AM to 1:40PM in Bloomberg Building 161X.

Students are required to submit.

- Final project presentations slides submitted as a PDF document
- Demonstration of deployed web applications

This document describes the final project presentation expectations and deliverables.

1. **Introduction (9 points)**

Motivation

- What problem are you tackling? (1 point)
- Why is it interesting/important? (1 point)
- What machine learning application did you develop and why is it useful for your target population? (2 points)

Technical focus

- What type of machine learning methods are used in the proposed approach and why is it well suited to address the problem. (2 points)
- What is particularly unique or novel about your approach? (1 point)

Impact:

- Discussion of social impacts and ethical concerns the application might raise. (1 point)

2. **Background (3 points)**

Student responses should include the following key points:

- Summarize prior work relevant to your project (1 point)
- Discuss how your proposed work is similar and different from prior work (1 point)
- Cite relevant sources (1 point)

3. **End-to-End ML Pipeline**

3.1 **Data Collection, Exploration & Processing (8 points)**

Student responses should include the following key points:

- Describe the dataset used for training machine learning models including quantity of data and source of the dataset (e.g., Kaggle). (3 points)
- State and justify **data exploration** methods used to gain knowledge about the dataset including figures/plots such as Scatterplots, Histograms, Boxplots, Lineplots, or others. Include figures to justify these insights. (3 points)
- State the **preprocessing** techniques used to prepare the dataset for ML algorithms including data cleaning (e.g., handling missing data), feature encoding (one-hot, integer), feature normalization, correlations, outlier removal, dimensionality reduction, feature creation, discretization, binarization, and augmentation. (2 points)

3.2 **Methods and Model Training (5 points)**

Student responses should include the following key points:

- Describe 2 or more machine learning algorithms used to solve the problem of interest. (1 point)

- Discuss the problem formulation, relevant equations, and theoretical underpinnings. (2 points)

- Describe model inputs and outputs. (2 points)

3.3 **Model Evaluation (4 points)**

Student responses should include the following key points:

- Describe at least **three evaluation metrics** used to measure the success of solving the problem. (1 point)

- Describe the training and validation procedure including the train/test split. Make sure that the experimental setup is described in enough detail to reproduce your results. (1 point)

- Describe the experiments including fine-tuning model hyperparameters. (1 point)

- Discuss the methods used to avoid model overfitting and underfitting. (1 point)

3.4 **Results (6 points)**

- Summarize the experimental results including the best and worst performing algorithms for at least 3 evaluation metrics. (6 points)

3.5 **Model Deployment and Front-End (Streamlit) (5 points)**

- Describe the web-based application you developed using the Streamlit library. (1 point)

- Describe and justify how the deployed model will be chosen. The machine learning algorithms may have tradeoffs in performance in terms of the selected evaluation metrics. Provide details about how the model was selected given these trade-offs. (1 point)
- Describe the target population and how the application built is beneficial to them. (1 point)
- Show and describe the user interface (UI) layout including user inputs (using menus, buttons, etc) and interface outputs. Show a video or give a live demonstration of the application. (2 points)

3.6 **Conclusion (5 points)**

- Summarize the project goals, ML approach, and key experimental results. (3 points)
- Provide discussion of the implications of project findings for your target user population and society broadly. (1 point)
- State future plans for this project, applying lessons learned from model training and evaluation. (1 point)

**Total Points:** 45 points